

Proof: Let, $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r)$ be True.

- By the defn. of OR, atleast one of $(p \wedge q \wedge r)$, $(p \wedge q \wedge \neg r)$, $(\neg p \wedge q \wedge r)$ must be True.
- Let, $(p \wedge q \wedge r)$ be True. By the defn. of AND, p is True, q is True, r is True. By the defn. of negation, $\neg(\neg r)$ is False. By the double negation law, $\neg(\neg r) \equiv r$. $\therefore r$ is False. By the defn. of negation, $\neg q$ is False. By the defn. of AND, $(p \wedge \neg q)$ is False. By the defn. of ~~negation~~ AND, $(p \wedge \neg q) \wedge r$ is False. By the defn. of negation, $\neg p$ is False. By the defn. of AND, $(\neg p \wedge q) \wedge r$ is False.

If $(p \wedge q \wedge r)$ be True, then p is True, q is True, r is False and $(p \wedge q \wedge \neg r)$, $(\neg p \wedge q \wedge r)$ are both False.

- Let, $(p \wedge q \wedge \neg r)$ be True. By the defn. of AND, p is True, q is True, $\neg r$ is True, r is False. By the defn. of negation, $\neg(\neg q)$ is False. By the ~~double~~ double negation law, $\neg(\neg q) \equiv q$. $\therefore q$ is False. By the defn. of negation, $\neg p$ and $\neg r$ are both False. By the defn. of AND, $p \wedge q \wedge \neg r$ is False. By the defn. of AND, $\neg p \wedge q \wedge r$ is False.

If $(p \wedge q \wedge \neg r)$ be True, then p is True, q is False, r is True and $(p \wedge q \wedge r)$, $(\neg p \wedge q \wedge r)$ are both False.

- Let, $(\neg p \wedge q \wedge r)$ be True. By the defn. of AND, $\neg p$ is True, q is True, r is True. By the defn. of negation, $\neg(\neg p)$ is False. By the double-negation law, $\neg(\neg p) \equiv p$. $\therefore p$ is False. By the defn. of AND, $(p \wedge q \wedge r)$ and $(p \wedge q \wedge \neg r)$ are both False.

If $(\neg p \wedge q \wedge r)$ be True, then p is False, q is True and r is True and both $(p \wedge q \wedge r)$ and $(p \wedge q \wedge \neg r)$ are False.

So, only one of $(p \wedge q \wedge r)$, $(p \wedge q \wedge \neg r)$, $(\neg p \wedge q \wedge r)$ can be True at a time.